Multimodal Machine Learning Lab

Winter Semester 2026/2027

Niklas Deckers and Martin Potthast

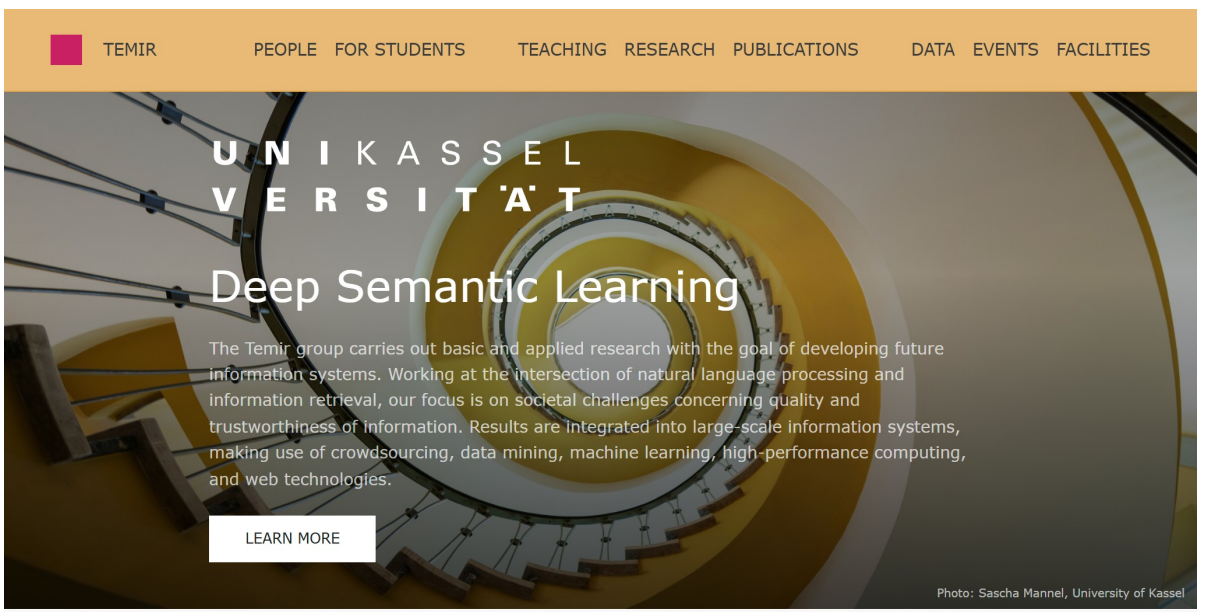
Deep Semantic Learning
University of Kassel and hessian.AI

About us





Martin Potthast Niklas Deckers



[kassel.webis.de]

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Agenda

- Learning and Research Objectives
- Lab Organization
- Getting to Know Each Other
- Motivation: Multimodal Models and CLIP
- Initial Task

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Learning Objectives

- Work in a structured and self-supervised manner
- Work on a project of a larger scope
- Deal with open-ended tasks
- Groupwork and communication skills
- Apply and extend current research and tools in the field of multimodal models
- Develop and carry out experiments
- Scientific writing
- Apply machine learning to a real life problem

We would like to do real research with you!

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Research Objectives for This Lab

- How can multimodal foundation models be used in specialized applications?
- What are the capabilities and weaknesses of such models and how could these be exploited?
- How can such weaknesses be mitigated?
- Objective: Develop prototypes for multimodal applications and demonstrate attacks on the underlying models
- Using modern tools to develop ML experiments (mode depends on number of participants)

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Lab Organization

- During the semester: Project work; weekly central sessions (online? depends on number of participants) + group consultations by individual appointment
- At the end of the semester: Group presentations and written reports
- 6 CP
- Material will be provided online

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Getting to Know Each Other

What experience do you have in the following subjects?

- Machine learning, deep learning

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- Prompt engineering

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Foundations

- Embeddings [mlvu.github.io]
- Word2Vec [temir.org]
- Multimodal models aim to align different modalities (text, image, audio, ...)
- CLIP (and generative models) [webis.de]
- Using CLIP as a foundation model (brainstorming: applications, vulnerabilities, mitigations)

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Initial Task

- **Until next week**
- Get on track with prerequisites: Using Python (and Pytorch), compute and compare CLIP embeddings of image–text pairs
- Search for application fields for CLIP: Find 5 exciting applications for CLIP (as precisely/actionable as possible; pairwise as different as possible)
- Later: Literature research and systematization of applications and vulnerabilities

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